RESEARCH REPORT

Recent Advancements in Biotechnology: Genetic Engineering (CRISPR, Gene Therapy)

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Table of Contents

Abstract

Biotechnology is rapidly advancing through developments in genomics, molecular biology, genome editing, gene delivery, and cell engineering. Genetic engineering has progressed from conventional gene-transfer methods toward increasingly programmable and precise manipulation of DNA. CRISPR-Cas systems have become among the most influential tools in this transformation because they can be directed to selected genetic sequences using guide RNA. Newer approaches, particularly base editing and prime editing, expand the range of genetic changes that can be introduced while reducing reliance on conventional double-strand DNA breaks. In parallel, gene therapy has moved from an experimental concept toward clinical application. Casgevy (exagamglogene autotemcel) represents a major milestone as the first FDA-approved CRISPR/Cas9-based therapy. In July 2026, its U.S. indication was expanded to patients aged 2 years and older with sickle cell disease with recurrent vaso-occlusive crises or transfusion-dependent beta-thalassemia. This report reviews major developments in genetic engineering, CRISPR technology, gene therapy, gene delivery, applications, challenges, ethical considerations, and future prospects.

Keywords: biotechnology; genetic engineering; CRISPR-Cas9; genome editing; gene therapy; base editing; prime editing

1. Introduction

Recent biotechnology is increasingly characterized by the ability to understand and manipulate biological systems at molecular resolution. Advances in high-throughput sequencing, functional genomics, bioinformatics, synthetic biology, and genome engineering have created new opportunities for research and therapeutic development. Genetic engineering has shifted from manipulating isolated genes toward programmable modification of specific genomic sequences. This is particularly important in medicine, where genetic engineering can be used to study disease mechanisms and develop interventions directed at their biological causes.

CRISPR-based genome editing is central to this transition. A guide RNA can direct a CRISPR-associated enzyme toward a selected nucleic-acid sequence. Conventional Cas9 editing produces a targeted DNA break that is repaired by cellular pathways. Later technologies have expanded the editing toolbox, allowing researchers to change individual bases, introduce small sequence changes, regulate gene expression, or explore larger genomic modifications. Gene therapy has developed alongside these technologies and now includes gene addition, gene replacement, gene silencing, genome editing, and ex vivo engineering of patient cells.

2. Objectives of the Study

The objectives of this report are to examine recent advances in genetic engineering; explain the principles and development of CRISPR-based genome editing; discuss emerging editing approaches; review progress in gene therapy and its clinical translation; describe applications in medicine, agriculture, and industrial biotechnology; and evaluate delivery barriers, safety concerns, ethical issues, and future prospects.

3. Genetic Engineering and the Evolution of Genome Editing

Modern genetic engineering includes altering nucleic-acid sequences, regulating gene expression, transferring genes between cells or organisms, and engineering cellular characteristics. Earlier recombinant-DNA methods enabled gene cloning and production of therapeutic proteins, but precise modification of endogenous genomic sites was more difficult. Programmable nucleases such as zinc-finger nucleases and transcription activator-like effector nucleases demonstrated that targeted genome modification was possible, although their design could be labor-intensive.

CRISPR changed this landscape because targeting can largely be redirected by changing the guide RNA rather than redesigning an entire protein for every genomic site. Cas9 therefore became a flexible platform for functional genomics, disease modeling, gene knockout studies, and therapeutic research. The field has expanded beyond Cas9 to include Cas12-family nucleases, RNA-targeting systems, epigenome editors, and other programmable tools. The overall development of genome engineering reflects a shift from simply cutting DNA toward writing, regulating, and restructuring genetic information.

4. CRISPR-Cas9 Technology

CRISPR is derived from an adaptive defense system found in many bacteria and archaea. In genome editing, guide RNA directs a CRISPR-associated nuclease to a complementary target sequence. Cas9 also requires recognition of an adjacent protospacer-adjacent motif (PAM). After target recognition, Cas9 can cut DNA, and cellular repair pathways can produce insertions or deletions or, under suitable conditions, incorporate a designed sequence.

5. Recent Advances in CRISPR Technology

5.1 Base Editing

Base editing allows certain nucleotide conversions without requiring a conventional double-strand DNA break. Cytosine and adenine base editors can perform specific transition changes. Because many disease-associated variants involve single-nucleotide substitutions, base editing has considerable therapeutic potential. Current research aims to improve editing efficiency, specificity, editing windows, and the range of possible nucleotide changes.

5.2 Prime Editing

Prime editing provides a broader editing strategy. It uses a Cas9-derived nickase linked to a reverse transcriptase and a specialized prime-editing guide RNA containing information for the desired change. The approach can potentially introduce substitutions, small insertions, and deletions without a conventional double-strand break or separate donor-DNA template. Important research priorities include improving efficiency, delivery, cell-type performance, and control of unintended outcomes.

5.3 Improved Nucleases and Other CRISPR Variants

Different CRISPR-associated proteins recognize different PAM sequences and can provide different targeting capabilities. Engineered nucleases are being developed to improve specificity and reach genomic sites that conventional Cas9 cannot efficiently access. CRISPR systems are also being adapted for transcriptional regulation, epigenetic modification, molecular diagnostics, and cellular recording.

5.4 Large-Scale DNA Engineering

Researchers are increasingly investigating ways to engineer larger genomic segments rather than individual bases or small sequence changes. Combining CRISPR targeting with recombination, transposition, or other DNA-writing mechanisms may eventually support multigene engineering or replacement of larger disease-associated regions. However, efficiency, specificity, delivery, and control remain major challenges.

6. Gene Therapy

Gene therapy aims to treat disease by changing genetic material or the behavior of genetically modified cells. It may involve adding or replacing a gene, silencing harmful genetic activity, editing a genome, or modifying cells outside the body before returning them to the patient. In vivo approaches deliver treatment directly into the patient, whereas ex vivo approaches remove cells, modify them in a controlled laboratory setting, and reinfuse them.

7. Casgevy and the Clinical Translation of CRISPR

Casgevy (exagamglogene autotemcel) is a landmark example of clinical genome editing. The therapy uses a patient's own hematopoietic stem cells, which are edited ex vivo using CRISPR/Cas9 and then returned to the patient after conditioning. The editing strategy disrupts a regulatory region associated with BCL11A in erythroid cells, increasing fetal hemoglobin (HbF). Increased HbF can reduce sickling of red blood cells in sickle cell disease and improve hemoglobin production in transfusion-dependent beta-thalassemia.

The FDA first approved Casgevy in December 2023 for eligible patients aged 12 years and older with sickle cell disease and recurrent vaso-occlusive crises, and it was subsequently approved for transfusion-dependent beta-thalassemia. In July 2026, the FDA expanded the indication to patients aged 2 years and older for both conditions. This development demonstrates the movement of CRISPR from laboratory research into patient care. However, treatment remains complex because it requires stem-cell collection, ex vivo manufacturing, conditioning, and reinfusion. Long-term monitoring is also important because genome editing can have unintended effects.

8. Applications of Genetic Engineering and CRISPR

Inherited disorders are major targets for genome editing because their causal variants can often be identified. Research and clinical development extend beyond sickle cell disease and beta-thalassemia to conditions such as hemophilia, muscular disorders, inherited retinal diseases, metabolic disorders, and immune deficiencies.

In cancer, CRISPR is used to investigate genes involved in tumor growth, drug resistance, and immune evasion. Genetic engineering also supports development of engineered immune cells, including chimeric antigen receptor (CAR) T-cell approaches designed to recognize cancer-associated targets.

Agricultural biotechnology uses genome editing to investigate disease resistance, stress tolerance, nutritional quality, and yield-related traits. Some genome-edited crops involve small targeted changes similar to naturally occurring variants. Regulation and public acceptance vary between countries.

In industrial biotechnology, engineered microorganisms and cell systems can produce enzymes, pharmaceuticals, biofuels, specialty chemicals, biomaterials, and food ingredients. CRISPR can accelerate metabolic engineering by allowing targeted modification of multiple genes and regulatory elements.

9. Advances in Gene Delivery

Delivery remains one of the major barriers to genetic medicine. Editing components must reach the correct cells at sufficient levels while limiting exposure of non-target tissues. Viral vectors such as adeno-associated virus (AAV) and lentiviral vectors remain important. AAV is widely investigated for in vivo delivery, while lentiviral vectors are commonly used for stable genetic modification of cells ex vivo.

10. Major Challenges and Limitations

Off-target editing and unintended on-target genomic changes are important safety concerns. Improved guide design, high-fidelity nucleases, engineered variants, and sensitive genomic analyses are being used to detect and reduce these risks. Delivery and tissue specificity are also difficult, especially for in vivo editing.

Long-term safety monitoring is essential because some genetic interventions can be durable or effectively permanent. Researchers must evaluate delayed adverse effects, clonal changes, immune complications, and unexpected genomic consequences. Immune responses to viral vectors, Cas proteins, or other therapeutic components may reduce efficacy or complicate repeat treatment.

Manufacturing and cost are additional barriers. Many gene and cell therapies require individualized processing, specialized facilities, strict quality control, and highly trained clinical teams. These requirements can increase costs and restrict access. Standardized manufacturing, improved logistics, and appropriate reimbursement systems are therefore important for wider implementation.

11. Ethical Considerations

Genome engineering raises ethical issues as well as scientific and clinical questions. Somatic editing changes non-reproductive cells and is generally distinguished from germline editing, in which alterations could be inherited by future generations. Germline editing raises particularly serious concerns because future individuals cannot consent to changes that may affect themselves and their descendants.

Other concerns include equitable access to expensive therapies, genetic privacy, potential discrimination based on genetic information, ownership and patent issues, enhancement beyond disease treatment, and possible misuse. Responsible development therefore requires informed consent, transparent evidence, independent oversight, long-term monitoring, and involvement of scientists, clinicians, regulators, patients, ethicists, and the public.

12. Future Prospects

The future of biotechnology is likely to integrate genome editing with stem-cell biology, synthetic biology, advanced delivery systems, computational biology, and personalized medicine. Base editing and prime editing may expand the number of genetic variants that can be addressed, while improved nucleases may increase targeting flexibility and specificity.

Artificial intelligence and computational methods may assist guide design, prediction of editing outcomes, identification of candidate targets, protein engineering, and analysis of large genomic datasets. Personalized gene therapy may also expand as sequencing and molecular diagnostics improve, allowing treatments to be selected according to a patient's specific genetic variant and cellular characteristics. Scientific progress, however, must be accompanied by improvements in safety, manufacturing, affordability, regulation, and global access.

13. Conclusion

Recent advancements in biotechnology have transformed genetic engineering into a powerful platform for therapeutic, agricultural, and industrial innovation. CRISPR-Cas9 has played a central role because of its programmability and versatility, while base editing, prime editing, improved nucleases, and large-scale DNA engineering are expanding the range of possible genetic interventions.

Gene therapy has entered an important stage of clinical maturity. Casgevy demonstrates that CRISPR-based genome editing can be translated into a clinical treatment, while its expanded FDA indication in 2026 illustrates continuing progress toward treatment of younger patients. Nevertheless, off-target and unintended on-target effects, delivery limitations, immune responses, long-term safety, manufacturing complexity, cost, and ethical concerns remain significant. The future of genetic engineering will therefore depend not only on making editing more powerful, but also on making it safer, more controllable, affordable, and equitable.

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